

Sharp but Wrong:

Velocity-Free Pitch-Control Values Are Not Calibrated Probabilities

An evaluation against 137,545 realised ball arrivals
179 matches, three competitions

September 17, 2026

Abstract

Pitch-control models return a number between zero and one for every point on the football pitch, and that number is routinely described, and used, as the probability that a team would control the ball if it arrived there. It is multiplied by positional values to produce expected possession value, integrated over regions to measure space creation, and compared against thresholds to recommend passes. Each of those operations assumes the number is a calibrated probability. We are not aware of any published test of that assumption, and the apparent reason is that control is unobserved almost everywhere, which makes the surface look unfalsifiable. It is not. Every pass, cross and clearance delivers the ball somewhere, and at that point the model's value is an ordinary probability forecast whose outcome is an ordinary draw. On 137,545 such arrivals from 179 matches, we find that velocity-free pitch-control values are not calibrated. A physics-based formulation discriminates well (AUC 0.762) yet earns a Brier score of 0.167, nearly twice as bad as forecasting the base rate everywhere (0.094); its calibration slope is 0.306, and a penalised logistic regression given exactly the same state beats it by 0.089 Brier (95% CI 0.082 to 0.096). The error is five times larger in the attacking third than the defensive third, grows monotonically with pass length, and is roughly seven times larger on the destinations players choose least often. It transports essentially unchanged across three competitions, including the men's-to-women's boundary, so it is structural rather than local. It is also almost entirely removable: a recalibration map fitted on two competitions strips 97–99% of it from a third. Downstream, aggregate possession value computed from raw control values is 28% lower than from calibrated ones, and 36% of threshold-based pass recommendations change. The ordering these models produce is sound; the numbers attached to that ordering are not.

Scope — stated before anything else

This paper evaluates the **velocity-free, camera-limited** variants of three pitch-control formulations on freeze-frame data. A freeze frame is a single snapshot, so no model here has access to player velocity, and the camera records a median of 16 of the 22 players. **This is not a test of published full-tracking pitch-control models.** It is a test of what those formulations do under the data conditions most clubs and researchers outside the elite tier actually work with — a different question, and one the results suggest is worth answering on its own terms. Section 10 returns to this at length.

All results are reproducible from `scripts/` on data obtainable with `scripts/fetch_data.py`. No number in this paper was chosen after seeing the data: the hypotheses, subgroups, horizons and metrics were fixed in the project's proposal beforehand, and the analysis order is frozen in code.

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1 A plain-language summary

This section is written for a reader who does not work in sports analytics. The rest of the paper assumes some statistics; this part does not.

What a pitch-control model does. Give it a photograph of a moment in a football match — where all twenty-two players are standing — and it colours in the whole pitch. Each point gets a number between 0 and 1. Red means “if the ball went here, the attacking team would end up with it”; blue means the defending team would. Panel (a) of Figure 1 is one of these pictures, computed from a real frame.

What the number is supposed to mean. If the model writes 0.7 on a patch of grass, and it is behaving like a probability, then across all the times it writes 0.7, the attacking team should end up with the ball about seven times in ten. Panel (b) of Figure 1 shows what that commitment looks like. This is a strong claim, and it is the one every downstream use of these models relies on: you cannot multiply a number by the value of a location, or add it up across a season, unless it is on a real probability scale.

Why nobody had checked. The colours cover the whole pitch, and the ball only goes to one place. Most of the picture describes something that never happened, so it looks untestable. But the part that *did* happen is testable: every pass lands somewhere. Take the model’s number at the spot where the ball actually arrived, then look at what actually happened next. Do that for 137,545 arrivals and you have an ordinary forecast-verification problem — the same problem weather forecasters solved half a century ago when they asked whether “70% chance of rain” meant anything.

What we found. The models are good at ranking and bad at scale. They reliably say a higher number for situations where the attacking team is more likely to keep the ball — that part works. But the numbers themselves are wrong, and wrong in a consistent direction. When the physics-based model says 0.4, the ball is in fact kept about eight times in ten (Figure 3). Its overall score is worse than if it had ignored the players entirely and guessed the same number every time (Figure 4). The error is worst exactly where these models get used most: in front of goal, and on long passes (Figure 5).

What it costs, and what fixes it. Because possession-value calculations multiply by these numbers, the error does not wash out — it compounds. Aggregate possession value comes out 28% too low, and about a third of “should I play this pass?” recommendations flip (Figure 8). The repair is mundane: one correction curve, fitted once on two competitions, removes 97–99% of the problem on a third (Figure 7). Two or three numbers, fitted once, and the field becomes usable for the arithmetic people were already doing with it.

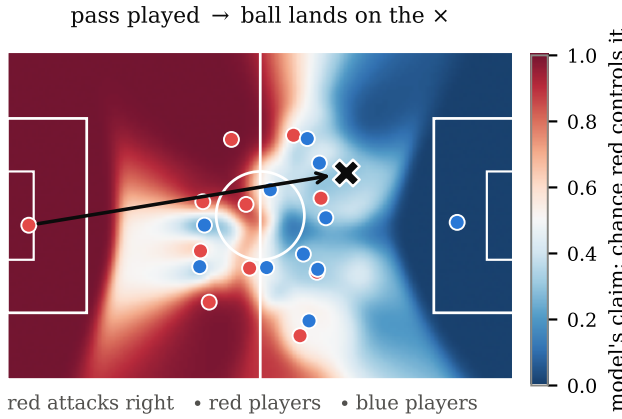
2 Introduction

A pitch-control model takes the instantaneous positions of the players and returns a field over the pitch,

$$C_A(x, y, t) \in [0, 1],$$

routinely described as the probability that team A would control the ball were it to arrive at point (x, y) at time t . The number is not merely displayed. It is multiplied by positional values

(a) What a pitch-control model outputs



(b) What that number would have to mean

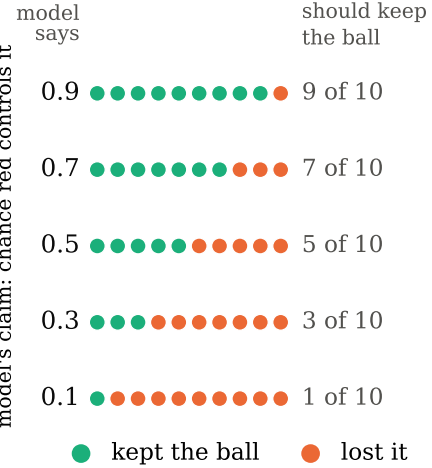


Figure 1. What a pitch-control model outputs, and what it would have to mean. (a) A control surface computed from a real freeze frame: red is the attacking team’s claim on each point, blue the defending team’s. The arrow is the pass that was actually played; the cross is where the ball arrived. Only the value at the cross is ever tested against an outcome. (b) What it means for the number to be a probability rather than a shading: across all the arrivals where the model says 0.7, the attacking team should keep the ball about seven times in ten.

to produce expected possession value, integrated over regions to produce space-creation metrics, differenced across counterfactual player positions to value off-ball runs, and compared against thresholds to recommend passes (Fernández and Bornn, 2018; Fernández et al., 2021; Spearman, 2018). Every one of those operations presupposes that C_A is a calibrated probability.

That presupposition has, as far as we can establish, not been tested. We are not aware of a published reliability diagram or proper-scoring-rule decomposition for any pitch-control model. The apparent obstacle is counterfactuality: control is unobserved at almost every point on the pitch, so the field looks unfalsifiable.

It is not. At every pass, cross and clearance the ball genuinely arrives somewhere, and at that point the model’s value is an ordinary probability forecast whose outcome is an ordinary draw. A season of data supplies tens of thousands of such forecast–outcome pairs. What follows is a standard probabilistic forecast evaluation, using apparatus settled in meteorology fifty years ago (Brier, 1950; Murphy, 1973) and not, to our knowledge, previously applied here.

Contributions

1. The first calibration assessment of pitch-control models we are aware of, on 137,545 arrivals across 179 matches and three competitions.
2. A demonstration that discrimination and calibration diverge sharply in this domain: the physics-based model ranks situations well (AUC 0.762) while scoring *worse than forecasting the base rate*.
3. A characterisation of where the error concentrates — the attacking third, long passes, poorly observed frames — and evidence that it is structural rather than local, transporting unchanged across competitions and across the men’s/women’s boundary.
4. Evidence that the defect is repairable: a recalibration map fitted on two competitions removes 97–99% of the measured miscalibration on a third.
5. Quantification of the downstream consequence: aggregate possession value is understated by

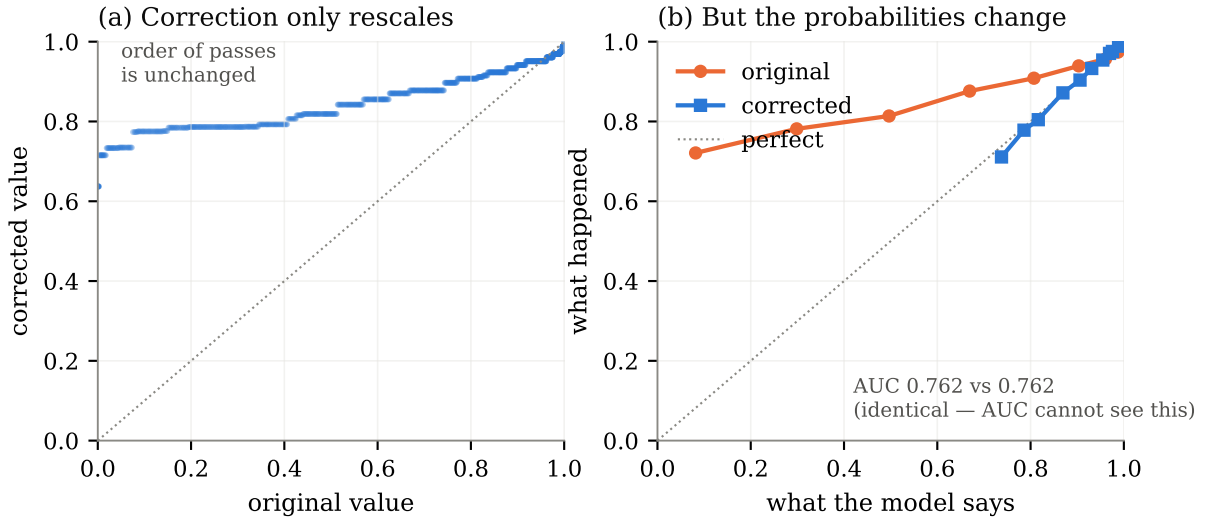


Figure 2. A correction that changes every probability and no rankings. (a) Each point is one arrival: its original forecast against its corrected one. The mapping is increasing everywhere, so no pass overtakes another. (b) The two versions plotted against what actually happened. The corrected forecasts sit on the diagonal, the originals far above it — and the AUC is identical (0.762 versus 0.762). A discrimination metric cannot see the difference between these two panels.

28%, and 36% of threshold-based pass recommendations change under recalibration.

6. A tested, open implementation of the whole protocol, including adapters for three providers and an instrument-validation suite.

3 Why a good AUC does not make a good probability

The single most important technical point in this paper is that two desirable properties of a forecast are independent, and the sports-analytics literature routinely reports only one of them.

Discrimination asks whether the forecast *orders* cases correctly: do the situations that turned out well receive higher numbers than the ones that turned out badly? The area under the ROC curve (AUC) measures exactly this and nothing else.

Calibration asks whether the numbers themselves are right: among all the cases assigned 0.7, did roughly 70% turn out well?

AUC is invariant to any strictly increasing transformation of the forecast. Replace every forecast p by p^3 , or by $0.5p + 0.25$, and the AUC does not move by a single decimal place — while the probabilities become arbitrarily wrong. Figure 2 makes this concrete on our own data: applying an isotonic recalibration map changes every forecast value substantially, leaves the ordering untouched, and leaves the AUC identical to three decimal places.

In plain terms. AUC is a test of who you rank first. Calibration is a test of what the number means. A model can win the first test and fail the second completely, and reporting only AUC is how a field ends up with numbers that look like probabilities and do not behave like them.

We take the standard position that a forecast should be *sharp subject to being calibrated* (Gneiting et al., 2007): calibration is the constraint, and informativeness is what one maximises within it. A forecast that is merely sharp — confident, spread towards 0 and 1 — is not thereby good, and the results below show a family of models that are very sharp and badly miscalibrated.

4 Related work

Dominant-region models partition the pitch by which player reaches each point first (Taki and Hasegawa, 2000; Fujimura and Sugihara, 2005). Physics-based formulations make arrival time uncertain and let control accumulate dynamically (Spearman et al., 2017; Spearman, 2018); related space-occupation and possession-value work embeds such surfaces in full valuation frameworks (Fernández and Bornn, 2018; Fernández et al., 2021). Brefeld et al. (2019) learn movement models from data and derive zones of control from them — the closest existing work to a statistically grounded control field. A broader survey of spatio-temporal football analysis is given by Gudmundsson and Horton (2017).

None of this literature reports whether the resulting values behave like probabilities. The apparatus for asking exists and is mature: proper scoring rules (Brier, 1950; Gneiting and Raftery, 2007), the sharpness-subject-to-calibration framing (Gneiting et al., 2007), score decompositions (Murphy, 1973), the calibration hierarchy (Van Calster et al., 2016, 2019), and binning-free reliability diagrams via isotonic regression (Dimitriadis et al., 2021). Recalibration is equally well established (Platt, 1999; Zadrozny and Elkan, 2002; Kull et al., 2017), as is the critique of binned calibration error (Nixon et al., 2019; Kumar et al., 2019; Vaicenavicius et al., 2019; Roelofs et al., 2022).

The contribution here is a transfer, not an invention. Full references, with unverified bibliographic details explicitly flagged, are in `docs/references.bib`.

5 Problem formulation

5.1 Three constructs, routinely conflated

Reachability (“who gets there first”) is a question about locomotion. Possession (“who ends up with the ball”) adds the contest at the point of contact. Retention (“who still has it after the next action”) adds the quality of the subsequent decision. These are different quantities, and in aggregate

$$C^{\text{reach}} \geq C^{\text{possess}} \geq C^{\text{retain}}.$$

Dominant-region models compute the first. The probabilistic reading used downstream asserts the second. We evaluate against the second, and return to the resulting objection in Section 10.

5.2 Operational definition of the outcome

The label

$Y = 1$ if and only if a player of team A is in possession at $t_{\text{arrival}} + h$, with $h = 1.0$ s. Arrivals whose horizon window is interrupted by a stoppage are **censored and excluded**, not counted as failures.

Censoring rather than coding stoppages as losses avoids biasing every model downward in the zones where fouls concentrate. It is not free: censoring is itself selective, and the censored fraction is reported throughout. Section 8.7 sweeps h and three alternative outcome definitions to show the conclusion does not depend on this choice.

5.3 The estimand

A model is calibrated on a population of arrivals if

$$\mathbb{E}[Y \mid C_A(X) = c] = c \quad \text{for almost every } c \text{ in the support of the forecast.}$$

This is *weak* calibration in the hierarchy of Van Calster et al. (2016) — conditioning on the forecast value, not on the full covariate vector. Strong calibration is not testable at realistic sample sizes and is not claimed.

5.4 Selection

The ball arrives only where someone sent it. Writing $D = 1$ for an observed arrival, the data identify $P(Y \mid X, D = 1)$ while the field claims to describe $P(Y \mid X)$. These coincide only if $Y \perp\!\!\!\perp D \mid X$. Two obstacles separate them: covariate shift in where the ball goes, which reweighting can in principle address (Horvitz and Thompson, 1952; Quiñonero-Candela et al., 2009), and selection on what the passer knew and the state vector does not contain, which it cannot. Section 8.5 treats both, and reports that the reweighting arm failed its own diagnostics.

6 Data

6.1 Corpus

We use 179 matches of StatsBomb 360 freeze-frame data across three competitions. 151,418 ball arrivals were extracted and 137,545 retained after the pre-registered inclusion criteria, of which 40,867 fall in the test fold across 53 held-out matches.

Table 1. The corpus. “Base rate” is the share of arrivals after which the attacking team is in possession one second later.

Competition	Matches	Arrivals	Base rate
FIFA World Cup 2022	64	51,390	0.909
UEFA Euro 2024	51	42,077	0.921
Women’s World Cup 2023	64	44,078	0.859

6.2 What the data does and does not contain

The following were verified against the files themselves rather than taken from documentation:

- **Flight time is measured**, not imputed: every pass carries a duration, with implied ball speeds of 6.9–22.1 m/s (median 13.1).
- **Possession is labelled at event level**, so the outcome is not reconstructed from a transfer log.
- **There is no velocity.** A freeze frame is one snapshot.
- **The camera limits what exists.** 360 frames cover 83.9% of passes; the median pass event shows 16 of 22 players; all 22 are visible at roughly 1 arrival in 2,500; and **21% of pass destinations fall outside the visible_area polygon**, where “no defender near the destination” means “no defender visible”.
- **No exogenous arrivals.** Only pass events carry both a start and an end location, so the corpus contains no deflections, clearances or second balls. This removes the strongest available identification strategy against selection (Section 8.5).

The derived outcome agrees with the provider’s own pass outcome on 96.6% of completed passes. On incomplete passes the two diverge — roughly half of failed passes still end with the

same team in control a second later — which is the construct distinction above made visible rather than an error.

6.3 Dependence

Arrivals cluster within matches. The measured intra-match ICC is 0.013 and the **design effect is 11.1**, so intervals computed as though arrivals were independent would be about 3.3 times too narrow. All intervals below are percentile cluster bootstraps over 1,000 replicates resampling whole matches (Cameron and Miller, 2015; Efron and Tibshirani, 1993).

I *In plain terms.* Two passes from the same match are not two independent pieces of evidence — the same teams, pitch and referee produced both. Treating them as independent would make our error bars roughly three times narrower than they should be, so we resample whole matches instead of individual passes.

7 Methods

7.1 Models

All models receive identical information and are evaluated on identical arrivals. The point of including fitted baselines is not to propose a better pitch-control model; it is to establish what is achievable from the same state vector, so that the unfitted models’ scores can be read against something other than zero.

Table 2. The six models. Only M0, M3 and M4 see the outcome during fitting; M2a fits a single scale parameter.

	Fitted?	Description
M0 Base rate	yes	Predicts the training base rate everywhere. Perfectly calibrated, useless; the reference for resolution.
M1 Voronoi	no	$C_A = \mathbf{1}[\min_j \tau_j < \min_k \tau_k]$. Hard 0/1 output.
M2 Physics	no	Logistic arrival times with competing-risks accumulation of control. Parameters $(\sigma, \lambda, v_{\max}, t_r)$ assumed, not estimated.
M2a Reachability sigmoid	one parameter	$C_A = \sigma((\min_k \tau_k - \min_j \tau_j)/s)$. Isolates how much of M2 is a monotone squash of a time-to-point difference.
M3 Logistic	yes	Penalised logistic regression on engineered kinematic features computed from exactly the state M1/M2 receive.
M4 GBM	yes	Histogram gradient boosting on the same features.

7.2 Evaluation

Splits are by **match**, never by row: arrivals nest within possessions within matches, and a random split rewards memorising a match’s shape. Recalibration maps are fitted on a validation fold disjoint from both the fitting and test data; the protocol raises an error if a split lacks one. A leakage audit is written with every run.

The primary calibration statistic is the **match-cross-fitted CORP miscalibration term** (MCB) of the Brier score (Dimitriadis et al., 2021), not binned expected calibration error. The CORP construction decomposes the Brier score as

$$\text{BS} = \text{MCB} - \text{DSC} + \text{UNC},$$

where MCB is the penalty for the probabilities being on the wrong scale, DSC is the credit for carrying information about which arrivals differ, and UNC is the irreducible difficulty of the question (the variance of the outcome). The reference forecast is obtained by pool-adjacent-violators isotonic regression rather than by binning.

We avoid binned ECE deliberately: it depends on an arbitrary bin count, is biased, is not a proper scoring rule, and can be driven near zero by coarsening (Nixon et al., 2019; Kumar et al., 2019). Our test suite constructs a forecast whose two-bin ECE is under 0.02 and whose forty-bin ECE exceeds 0.08. One caveat of the CORP estimator matters here and is handled explicitly: in sample, MCB is biased *upward*, because the isotonic fit absorbs noise. We therefore cross-fit by match, and we do not truncate negative estimates.

I *In plain terms.* MCB is “how much of the score is wasted on the numbers being on the wrong scale”. Zero means the probabilities mean what they say. It is the single number this paper is about, and throughout the figures it is labelled simply “size of the error”.

7.3 Instrument validation

Before measuring anything, the calibration estimator was validated against a known ground truth on simulated data. It reports near-zero miscalibration for an oracle forecast; recovers an injected logit distortion to within 0.01 of the injected slope; confirms that AUC is unchanged to machine precision by that distortion; and produces intervals 2.3 times wider under clustering than a naive bootstrap. All eight checks pass and are re-run by `scripts/09_validate_instrument.py`.

This step is worth naming explicitly. A calibration finding is a claim about an instrument as much as about a model, and an uncalibrated instrument would produce exactly the paper we did not want to write.

8 Results

Test fold: 40,867 arrivals across 53 held-out matches, base rate 0.894.

8.1 Overall calibration

Table 3. Headline results on the held-out fold. Brackets are 95% percentile cluster-bootstrap intervals over 1,000 match resamples. Forecast SD is the standard deviation of the model’s output — a measure of how confident it is.

Model	Brier	95% CI	MCB	DSC	Slope	AUC	Forecast SD
M1 Voronoi	0.231	[0.216, 0.244]	0.142	0.006	0.114	0.675	0.428
M2 Physics	0.167	[0.156, 0.177]	0.081	0.008	0.306	0.762	0.313
M2a Reach. sigmoid	0.155	[0.146, 0.163]	0.069	0.008	0.636	0.761	0.247
M0 Base rate	0.094	[0.088, 0.101]	0.000	0.000	—	0.500	0.000
M3 Logistic	0.078	[0.072, 0.084]	0.001	0.017	1.014	0.837	0.133
M4 GBM	0.073	[0.068, 0.079]	0.000	0.021	1.018	0.864	0.145

The MCB interval for M2 is [0.075, 0.087]; for M1, [0.133, 0.152]; for M2a, [0.064, 0.073]. None of them contains zero.

The unfitted geometric models are not calibrated. The physics model’s calibration slope of 0.306 is the signature of severe overconfidence in scale: the forecast spreads towards 0 and 1

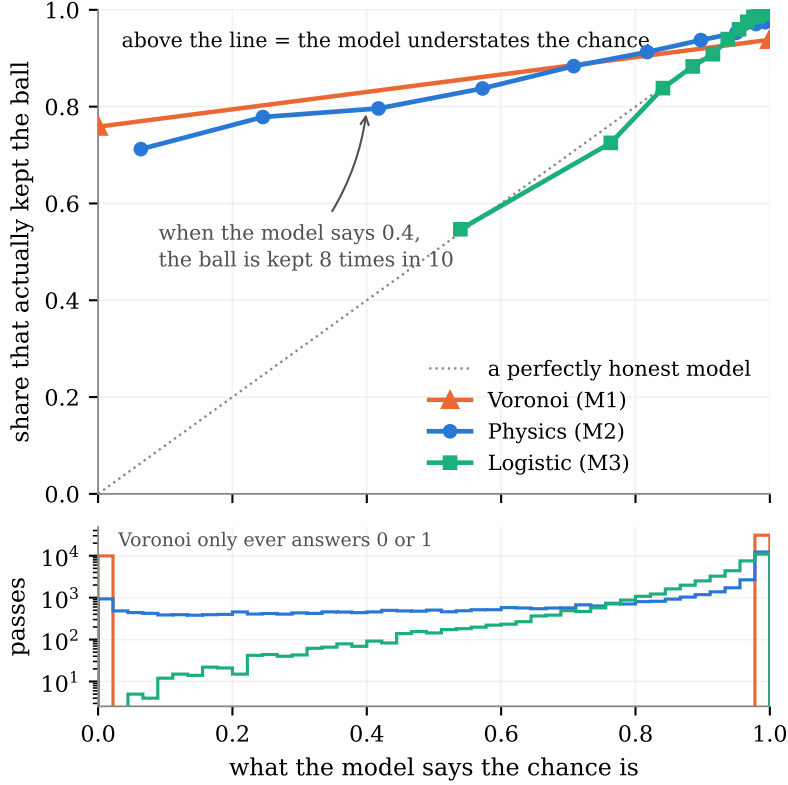


Figure 3. Reliability of three models on the held-out fold. A perfectly honest model would trace the dotted diagonal. The two unfitted models sit well above it — they understate the attacking team’s chances by 0.14 (Voronoi) and 0.17 (physics) on average — while the fitted logistic lies on it to within 0.001. The lower panel shows how often each value is actually emitted; the physics model spends most of its output range in a region where its numbers are badly wrong, and the Voronoi model emits nothing but 0 and 1.

roughly three times further than the evidence supports. Its mean forecast is 0.72 against a base rate of 0.89, so it also under-forecasts systematically. Figure 3 shows this directly.

The central observation of this paper

M2 discriminates well — AUC 0.762, far above chance — while scoring 0.167 on the Brier score, nearly twice as bad as forecasting the base rate everywhere (0.094). **A model that looks good on a discrimination metric can be worse than useless as a probability.** M0 makes the converse point: it is perfectly calibrated (MCB 0.000) and carries no information whatever (DSC 0.000, AUC 0.500). Neither property substitutes for the other.

Figure 4 shows where each model’s score goes. The grey segment is the difficulty of the question, identical for every model; the green segment is the credit a model earns for being informative; the orange segment is the penalty it pays for being on the wrong scale. For M1, M2 and M2a the penalty dwarfs the credit by an order of magnitude.

8.2 Comparison against fitted baselines

Table 4 gives paired cluster-bootstrap differences in Brier score against M3, computed on identical test arrivals. Positive values favour the baseline.

A penalised logistic regression, given *exactly* the state the physics model receives, is decisively better. Two further observations follow. M2 and M2a differ by 0.012 Brier while both sit 0.08–0.09 above M3, so the competing-risks dynamics add little beyond a monotone squash of

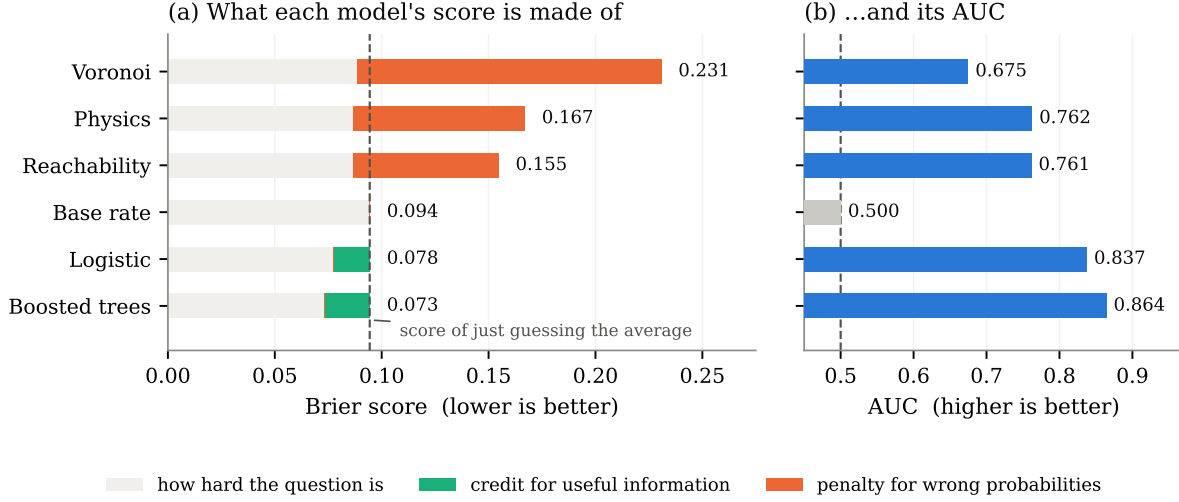


Figure 4. Where each model’s score comes from, and how it looks on AUC. (a) The Brier score decomposed: difficulty of the question, minus credit for useful information, plus penalty for wrong probabilities. Everything to the right of the dashed line is worse than guessing the average. (b) The same models ranked by AUC. The physics and reachability models sit comfortably above chance in (b) while being the second- and third-worst forecasts in (a).

Table 4. Paired differences in Brier score against the penalised logistic baseline (M3). “Baseline” means the baseline wins.

Model	Δ Brier vs M3 [95% CI]	Verdict
M1 Voronoi	+0.153 [0.142, 0.163]	baseline
M2 Physics	+0.089 [0.082, 0.096]	baseline
M2a Reach. sigmoid	+0.077 [0.072, 0.082]	baseline
M0 Base rate	+0.017 [0.015, 0.019]	baseline
M4 GBM	−0.005 [−0.005, −0.004]	model

a time-to-point difference — most of what M2 does could be done with one fitted parameter. And M2’s DSC (0.008) is less than half M3’s (0.017), so the gap is not only scale: the physics model also carries less information about which arrivals differ.

8.3 Where the error concentrates

All 47 pre-registered strata show FDR-adjusted evidence of miscalibration ($\alpha = 0.05$). Every directional prediction registered in advance is supported.

Miscalibration is five times worse in the attacking third than the defensive third, rises monotonically with pass length, and is worst of all in the central attacking fifth — the penalty area, where the mean forecast is 0.34 against an observed rate of 0.84. This is precisely the region where space-creation and possession-value metrics are most used and most consequential. Figure 5 maps it.

8.4 How much of it is the camera?

Calibration improves monotonically with the fraction of the frame observed (Table 6, Figure 6). Destinations outside the visible area are far worse (MCB 0.145, slope 0.167, $n = 8,565$) than those inside it (0.064, 0.390, $n = 32,302$).

Table 5. Selected strata for the physics model (M2). MCB is the cross-fitted miscalibration term; slope is the calibration slope, where 1.0 would be honest.

Stratum	n	MCB	Slope
Attacking third	12,161	0.158	0.273
Middle third	20,448	0.057	0.359
Defensive third	8,258	0.031	0.324
Pass 0–10 m	10,790	0.070	0.447
Pass 20–30 m	7,484	0.098	0.264
Pass 30–45 m	3,124	0.126	0.190
Pass 45 m+	987	0.156	0.092
Pressure: 0 opponents	12,679	0.075	0.306
Pressure: 3+ opponents	1,705	0.114	0.336
Attacking fifth, central	2,175	0.292	0.185

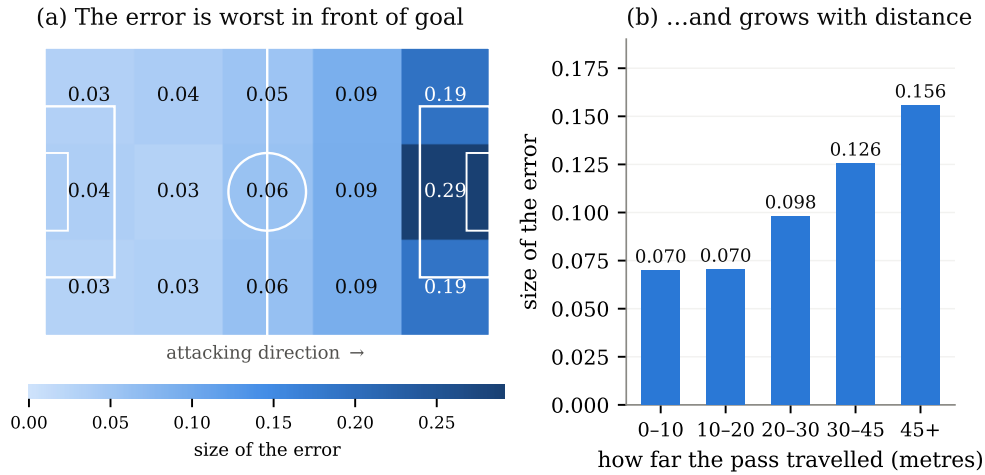


Figure 5. The error is not spread evenly. (a) Miscalibration by pitch zone for the physics model, attacking left to right. The worst cell is the central attacking fifth — the penalty area — at nearly ten times the error of the defensive flanks. (b) The same statistic by pass length: it more than doubles from short passes to long ones.

Restricting to near-complete frames *with* a visible destination gives MCB 0.045 and slope 0.571, against 0.081 and 0.306 overall — so roughly **45% of the measured miscalibration is attributable to incomplete observation and 55% survives it**. Even when the camera sees nearly everything, the velocity-free model remains badly overconfident while its AUC rises to 0.86: the same divergence, at better data quality.

This decomposition is **suggestive, not causal**. High-completeness frames are not a random subsample — the ball is more often in crowded central areas — so the comparison confounds observability with the kind of situation observed.

Table 6. Miscalibration of the physics model by how much of the frame the camera recorded.

Frame completeness	n	MCB	Slope
< 60%	11,040	0.095	0.221
60–75%	12,277	0.086	0.300
75–90%	13,104	0.076	0.385
90%+	4,446	0.049	0.556

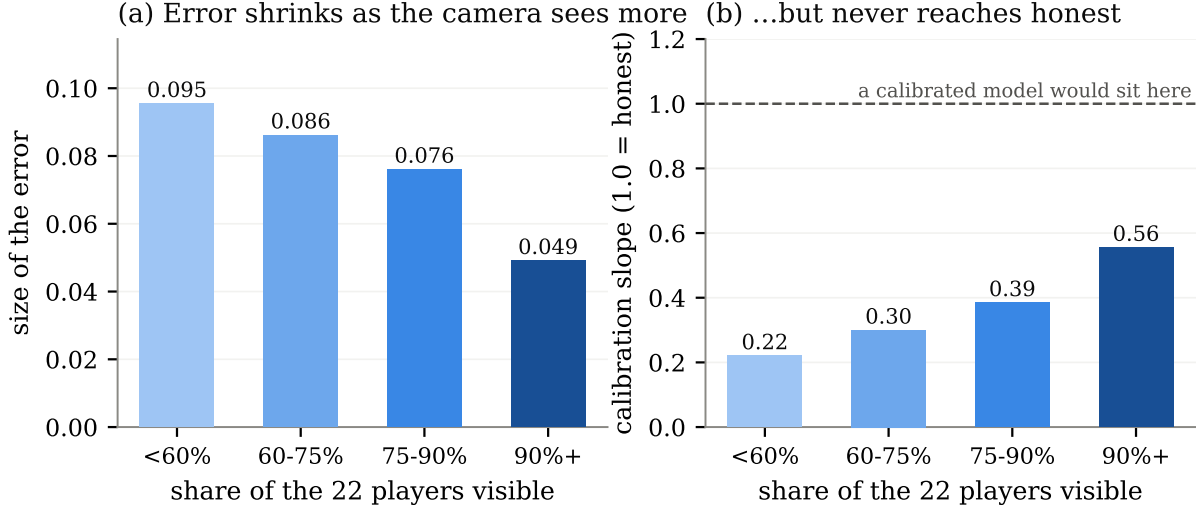


Figure 6. Better observation helps, and is not enough. (a) The error falls by roughly half as the share of visible players rises. (b) The calibration slope rises towards 1.0 but stops at 0.56 even in the best-observed frames — values below 1.0 mean the probabilities are spread too far towards 0 and 1. Note that better-observed frames are not a random subsample, so this is an association, not a causal decomposition.

8.5 Selection

A negative result, reported as one

The reweighting arm of the analysis **failed its own diagnostics, and we report it as failed rather than reporting the number it produced.**

A classifier separating chosen from proposed destinations achieves AUC 0.972: passers select their targets very strongly, which confirms the premise that selection is a real threat. But the overlap diagnostics are damning. **52.6% of observed arrivals lie outside the support of the candidate distribution**; the top 1% of weights carry 33% of the mass; the largest weight is 521; and the effective sample size collapses from 48,268 to 7,753 (16%) under the trimmed, stabilised weights the analysis actually uses — 1,525, or 3%, before trimming. Under those conditions the reweighted estimate (MCB 0.127 against an unweighted 0.085) is extrapolation, not correction, and we do not interpret it.

Stratification is more informative. Table 7 splits the test fold by quintile of the selection score, where stratum 0 holds the destinations a passer would be least likely to choose.

The model is worst on the destinations players choose least often, by a factor of roughly seven. The headline figure of 0.081 is therefore a weighted average dominated by routine passes, and it *understates* the error in the unusual situations where a control model would most plausibly be consulted.

The strongest available identification strategy could not be run. The proposal’s primary answer to selection was a quasi-exogenous subsample — deflections, clearances and

Table 7. Miscalibration by how typical the destination is. The model is worst exactly where a practitioner would most want to consult it.

Stratum	n	Base rate	Mean forecast	MCB
0 (least typical)	9,654	0.790	0.507	0.148
1	9,653	0.896	0.612	0.157
2	9,654	0.932	0.782	0.065
3	9,653	0.959	0.865	0.039
4 (most typical)	9,654	0.969	0.906	0.022

second balls, which arrive where nobody aimed. This corpus contains none: only pass events carry both a start and an end location. That is a real limitation, and its absence is why the sensitivity analysis carries more weight here than intended.

That sensitivity analysis is reassuring. Modelling unmeasured selection as a multiplicative bias Γ on the outcome odds among observed arrivals, in the style of Rosenbaum and Rubin (1983), MCB remains 0.039 even at $\Gamma = 3$ — an implausibly strong assumption about what passers know and the state vector does not. **No tipping point exists within the range examined:** the finding is not explicable by unmeasured selection of any plausible magnitude.

8.6 Does it transport, and can it be fixed?

Table 8. Leave-one-competition-out transportability. “Penalty for crossing” is the difference between the two preceding columns.

Model	MCB within competition	MCB across competitions	Penalty for crossing
M2 Physics	0.085	0.083	− 0.002
M3 Logistic	0.0010	0.0011	+0.0001
M4 GBM	0.0006	0.0004	−0.0002

Crossing a competition boundary costs essentially nothing, **including the men’s-to-women’s boundary**, despite base rates differing materially (0.909, 0.921, 0.859). The miscalibration is structural, not local: the model is equally wrong everywhere. That is a stronger and more useful claim than being wrong on one league.

It follows that one fix travels. A recalibration map fitted on two competitions and applied to a third, unseen one removes **97–99%** of the physics model’s measured miscalibration (isotonic 98.9% on average, worst case 98.6%; beta 98.6%/98.3%; Platt 97.4%/96.4%). On the main match-level split, all three maps drive MCB from 0.081 to within 0.001 of zero. Figure 7 shows both halves of the result.

The model’s *ordering* of situations is therefore sound; only its scale is wrong. The defect is repairable downstream by a two- or three-parameter map that a practitioner can fit once and deploy — a deflating result, and a useful one.

8.7 Does the conclusion survive the definition of control?

The study imposes an operational definition on an undefined construct. The pre-registered rule was that a conclusion which flips across the sweep would be reported as flipping. It does not flip.

Across six horizons and four outcome definitions, MCB stays in [0.064, 0.088] and the slope in [0.219, 0.306]. The conclusion is not an artefact of the horizon. Notably, the **first-touch**

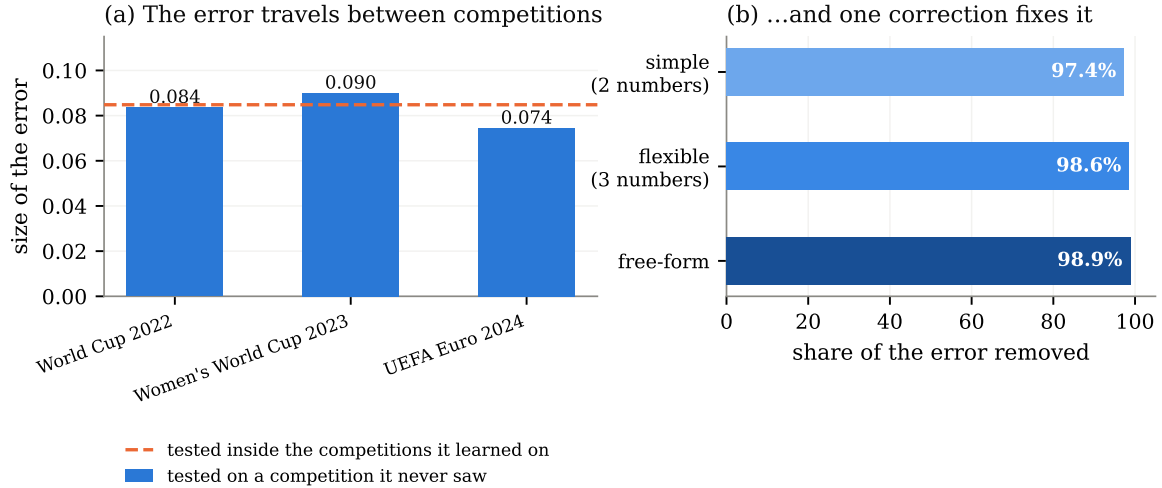


Figure 7. The defect travels, and so does the repair. (a) Holding out each competition in turn changes the error hardly at all relative to testing inside the competitions the map was built on — including across the men’s/women’s boundary. (b) A correction fitted on two competitions and applied to the third removes almost all of the error, and a two-parameter map is nearly as good as a free-form one.

Table 9. Sensitivity to the outcome definition and horizon, physics model (M2).

Definition	Horizon	Base rate	Censored	MCB	Slope	AUC
Control at horizon	0.25 s	0.917	0.6%	0.088	0.306	0.772
Control at horizon	1.0 s	0.911	1.2%	0.084	0.300	0.768
Control at horizon	3.0 s	0.883	2.9%	0.070	0.278	0.748
First touch	—	0.918	0.0%	0.088	0.304	0.772
Retained sequence	—	0.912	1.2%	0.084	0.303	0.770
Possession at 5 s	—	0.855	1.9%	0.064	0.219	0.705

definition — the closest observable proxy for the reachability construct that dominant-region models actually compute — gives MCB 0.088, no better than the possession definition. The models are not merely being scored against the wrong target.

Censoring rises from 0.6% to 2.9% as the horizon lengthens. Since fouls cluster in contested areas, a longer horizon buys construct breadth at the cost of a more selected sample; the effect is small at these rates but is reported rather than assumed away.

9 Practical implications

9.1 Possession value is understated by more than a quarter

Propagating raw and recalibrated control values through a linear expected-possession-value construction over the 40,867 test arrivals gives Table 10.

Aggregate possession value computed from raw control values is **28% lower** than from calibrated ones. This is the accumulation argument made concrete: because EPV is linear in C , a calibration error with a consistent sign does not average out across a season — it compounds.

Table 10. Expected possession value computed from raw and recalibrated control values, at a loss discount of 0.35.

Control variant	Mean EPV	Total EPV
Raw	0.559	22,844
Platt	0.774	31,613
Isotonic	0.773	31,581
Beta	0.773	31,576

Table 11. Effect of recalibration on threshold-based pass recommendations.

Threshold	Recommendations changed	Accuracy raw	Accuracy recalibrated
0.50	25.2%	0.762	0.895
0.60	30.6%	0.727	0.895
0.70	36.0%	0.680	0.892
0.80	24.0%	0.622	0.792

9.2 A third of pass recommendations change

9.3 But read the mechanism carefully

Recalibration improves net benefit (Vickers and Elkin, 2006) at 90 of 91 thresholds, and doubles the range over which the model beats the best default policy (from 15 thresholds to 31). It is tempting to conclude that recalibration makes the model better at football. It does not.

Every map used here is monotone, so AUC is unchanged to machine precision: the model’s *ranking* of arrivals is identical before and after. What recalibration restores is the **meaning of the threshold**. A practitioner who sets a cut-off from a stated risk appetite — “attempt the pass when control probability exceeds 0.7” — is, with the raw forecast, actually operating at a far stricter point than intended, because a raw 0.7 corresponds to an observed rate near 0.9. A practitioner who instead tuned their threshold empirically on the raw forecasts would already be getting the same decisions.

So the honest statement is narrower than “recalibration improves decisions”:

The correct reading

Miscalibration corrupts any use of the value that is not purely ordinal — summing, multiplying, integrating, or comparing against an externally chosen threshold. Purely ordinal uses, such as ranking a player’s passing options against each other, are unaffected.

10 Limitations

This does not test full-tracking pitch-control models. The single most important limitation. No model here has velocity, and velocity is the primary input the physics formulations were designed around. The results bound what those models do under degraded data; they do not measure what they do under the data they were built for. The observability analysis suggests observability explains under half of the error, but that decomposition is not causal, and the velocity component cannot be separated at all on this corpus.

Construct mismatch is a live objection. Dominant-region models answer “who arrives first”; we score against “who ends up with the ball”. A defender of these models could argue we

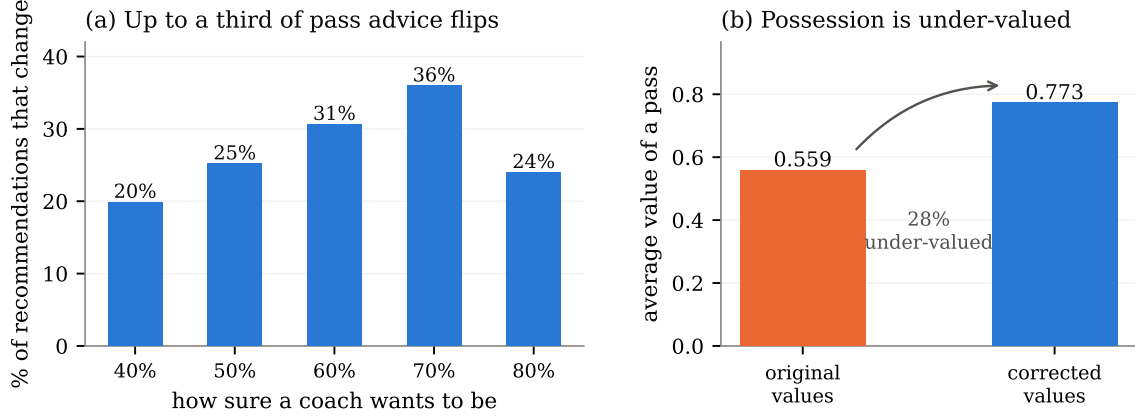


Figure 8. What the miscalibration costs in football terms. (a) The share of pass recommendations that change once the control values are corrected, as a function of how sure a coach wants to be before playing the pass. (b) Aggregate possession value over the test fold, before and after correction.

are scoring them against a target they never claimed. Two responses. First, the downstream literature does use these fields for the second question; if they answer only the first, that invalidates the downstream uses rather than excusing the models. Second, and more directly: under the first-touch definition, the closest observable proxy for reachability, MCB is 0.088 — no better. The mismatch does not explain the result.

The selection analysis is incomplete. The quasi-exogenous subsample could not be constructed, the reweighting failed its overlap diagnostics, and we are left with stratification and a sensitivity bound. The bound is strong (no tipping point below $\Gamma = 3$), but it is a bound, not identification.

Observational data cannot identify control where the ball never arrives. Our conclusions extend to realised arrivals. That is also where every downstream application operates, so the limitation is narrower than it sounds — but the full field’s claims remain untested and, on observational data, untestable.

Three competitions, two of them elite men’s tournaments. Transportability held across the men’s/women’s boundary, which is evidence against population-specificity, but youth and lower-division football are untested and the locomotion assumptions have less reason to hold there.

Parameter sensitivity was not exhausted. The arrival-time uncertainty σ governs the field’s sharpness directly, and our ablations on smaller corpora showed it moving the calibration slope by more than the difference between models. A σ fitted to outcomes would likely improve M2 substantially — which is itself the point: a parameter that must be fitted to outcomes is not physics.

11 Conclusion

On 137,545 realised ball arrivals, velocity-free pitch-control values are not calibrated probabilities. The physics-based model scores worse than forecasting the base rate while discriminating well

above chance; its calibration slope is 0.306; a penalised logistic regression given identical inputs is decisively better. The error concentrates in the attacking third, grows with pass length, worsens where the camera sees less, and is seven times larger on the destinations players choose least often. It transports unchanged across three competitions and across the men’s/women’s boundary, and it survives every horizon and outcome definition we pre-registered.

It is also almost entirely removable. A recalibration map fitted on two competitions strips 97–99% of it from a third. The ordering these models produce is sound; the numbers attached to that ordering are not.

The practical recommendation is therefore specific rather than dismissive. Pitch-control fields remain useful for ranking. They should not be summed, integrated, multiplied by positional values, or compared against externally chosen thresholds without recalibration — and on this evidence a single global map suffices to make them fit for those uses.

Whether the same holds for full-tracking implementations is the obvious next question, and it needs a corpus this study did not have.

A Reproduction

```

pip install -r requirements.txt && pip install -e .
python scripts/09_validate_instrument.py          # validate the estimator first

for spec in "43 106" "55 282" "72 107"; do set -- $spec
  python scripts/fetch_data.py --source statsbomb_open --accept-terms \
    --competition $1 --season $2 --with-360
done

SRC="--source statsbomb_open"

python scripts/03_main_analysis.py                $SRC --n-boot 1000
python scripts/04_subgroup_analysis.py            $SRC
python scripts/05_selection_analysis.py           $SRC --max-matches 60
python scripts/06_downstream_analysis.py          $SRC
python scripts/11_transportability.py              $SRC
python scripts/12_construct_sensitivity.py         $SRC --max-matches 64
python scripts/14_make_figures.py                 # rebuild every figure

```

Every results directory carries a manifest recording the git revision, package versions, and a hash of the configuration. Protocol, hypotheses and subgroups are in `docs/proposal.md`; dataset verification in `docs/data_sources.md`. `scripts/13_verify_paper.py` re-checks every numeric claim in this paper against the files in `results/`.

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